MIL-B-40028B
12 April 1976
SUPERSEDING
MIL-B-40028A
29 March 1968

MILITARY SPECIFICATION

BAGS, BARRIER, WITH VOLATILE CORROSION

INHIBITOR TREATED LINERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers opaque, heat sealable, water vaporproof barrier bags, lined with a volatile corrosion inhibitor (VCI) treated paper.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal		
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-665	-	Boxes: Paperboard, Metal Edged and Components
Military		
MIL-P-116	-	Preservation-Packaging, Methods of
MIL-B-131	-	Barrier Materials, Watervaporproof, Flexible, Heat-Sealable
MIL-P-3420	-	Packaging Materials, Volatile Corrosion Inhibitor Treated, Opaque
MIL-I-8574	-	Inhibitors, Corrosion, Volatile, Utilization of

FSC 8105

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.

MIL-P-14232 - Parts, Equipment and Tools for Army Materiel, Packaging and Packing of

STANDARDS

Federal

Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procurring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

ASTM D1583 - Hydrogen Ion Concentration of Dry Adhesive Films, Test for

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2).

3.2 Materials.

3.2.1 Bags. Bags shall be manufactured from barrier material conforming to Class 1 of MIL-B-131, with a volatile corrosion inhibitor (VCI) treated liner as specified in 3.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.

3.2.2 Bag liner. Unless otherwise specified (see 6.2), liner material shall be VCI treated Type I, Class 3, Style A in accordance with MIL-P-3420. Application and handling of VCI treated material shall be in accordance with MIL-I-8574.

3.2.3 Adhesive. The adhesive used to secure the liners and reinforcing patches shall have a hydrogen ion concentration (pH) of 6.0 to 8.5. The adhesive shall be dry to the touch in 48 hours. Testing shall be as specified in 4.5.4.

3.3 Construction.

3.3.1 Bags. Bags shall be constructed as specified on the applicable drawings or as specified in the contract (see 6.2).

3.3.1.1 Reinforcing patches. Barrier materials, as specified in 3.2.1, shall be used as reinforcing patches when required. The reinforcing patches shall be cemented in place with adhesive (see 3.2.3) in accordance with the applicable drawing.

3.3.2 Liners. The untreated side of the liner shall be spot tacked to the bag in accordance with the applicable drawing. The liner shall be positioned in the bag to insure its opening at the same time as the bag. Liner surfaces shall not adhere to each other.

3.3.3 Dimensions. The dimensions of the bag and liner shall be in accordance with the applicable drawing or as specified in the contract (see 6.2).

3.4 Performance characteristics.

3.4.1 Water resistance. The seams of the finished bag shall be water resistant. Testing shall be as specified in 4.5.4.

3.4.2 Seam strength. The seam strength shall be in accordance with MIL-P-116. Testing shall be as specified in 4.5.4.

3.5 Marking. The VCI lined bags shall be individually marked with the manufacturer's name, date of manufacture (month and year) and this specification number. The letters and figures shall be a minimum of 0.12 inch high, legible and water-resistant. The marking shall appear on the outside of the bag.

3.6 Workmanship. The bags shall be free from sharp creases, wrinkles, and other imperfections which may be detrimental to their function.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The contractor shall perform, as a minimum, inspections in accordance with the specifications, Quality Assurance provisions and the contract. These minimum inspections shall not be construed as relieving the contractor of his responsibilities under terms of the contract to furnish the Government with items complying with and conforming to the requirements of the product drawings and specifications.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.2). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein and such other inspection as necessary to determine that all requirements of the contract have been met.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified (see 6.2), the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105 and shall consist of not more than 10,000 bags. When Acceptable Quality Levels (AQL's) are specified, the AQL's shall be individually applied in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.

4.5.2 Certification provisions.

4.5.2.1 Certified test reports (CTR). When specified in the contract or in documents referenced therein, the contractor shall make available to the Government a certified test report for each lot of items by lot number prior to acceptance. This test report is in addition to, and not in lieu of, any rights of the Government under this contract or law. A CTR may be used as an element incident to, but shall not be used as the sole basis for, Government acceptance of the contract item(s). As a minimum, the report shall contain the following:

a. Name of company and date.
b. Contract number or purchase order number, national stock number and drawing number.
c. Complete nomenclature of supplies together with lot number or other identification. The quantity in each lot or shipment shall be given.
d. All inspections and tests required by contract (i.e., material, processes, performance, functional, etc.) shall be recorded in test reports. These reports shall identify each lot, submitted for acceptance by lot number, the specification or drawing, revision and date, grade or type as applicable, number of specimens tested, specified characteristics and requirements, and actual results obtained.
e. Reports of the raw material producer's chemical, mechanical and physical analyses.
f. A statement, as follows, certifying that material meets all requirements of the contract:

"The undersigned individually, and as the authorized representative of the contractor, warrants and represents that: All of the information supplied above is true and accurate; the material covered by this certificate conforms to all contract requirements (including but not limited to the drawings and specifications); the inspection and test results, and the analyses appearing herein are true and accurate; and this certificate is made for the purpose of inducing payment and with knowledge that the information and certification may be used as a basis for payment."

g. Signature and title of certifying official.

4.5.3 Examination. A random sample of bags and liners shall be taken from each lot in accordance with MIL-STD-105, Inspection Level I, and examined in accordance with Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.

TABLE I

Characteristics	AQL	Inspection Method
1. Materials (3.2)	—	Certification (see 4.5.3)
2. Damaged barriers (tears, cuts, chafed spots)	0.65	Visual*
3. Damaged barriers (tears, cuts, delamination)	0.65	Visual*
4. Missing liners or reinforcing patches	0.65	Visual*
5. Treated VCI surface facing barrier material	0.65	Visual*
6. Adhesion of liners to each other (see 3.3.2)	0.65	Visual*
7. Position of liner (see 3.3.2)	4.0	Visual
8. Inside dimensions (see 3.3.3)	4.0	Visual
9. Markings: Missing, illegible or incorrect (see 3.5)	4.0	Visual
10. Workmanship (see 3.6)	4.0	Visual

* Simultaneous inspection for liners adhering to each other and undersize dimensions may be accomplished by inserting a template into the assembled bags. The template may be fabricated from heavy fiberboard or other suitable material and made to the minimum allowable dimensions specified for the inside of the bags.

4.5.4 Testing. The end item shall be tested in accordance with Table II. A double sampling plan of MIL-STD-105 shall be used. The Inspection Level shall be S-4 with an AQL of 2.5 percent defective.

TABLE II

Characteristics	Requirement	Test Method
Water resistance	3.4.1	4.6.1
Seam Strength	3.4.2	4.6.2
Adhesive	3.2.3	4.6.3

4.5.5 Inspection of packaging. Unless otherwise specified, (see 6.2) inspection to determine compliance with preservation, packing and marking requirements of the applicable packaging documentation, for the level designated in the contract, shall be as specified in MIL-P-14232.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.

4.6 Test methods.

4.6.1 Water resistance. The water resistance test shall be performed in accordance with the vacuum chamber or submersion technique specified in MIL-P-116. The sealed bag shall contain shredded paper or other material to provide for the entrapment of air.

4.6.2 Seam strength. The seam strength test shall be performed in accordance with the heat-sealed seam test specified in MIL-P-116.

4.6.3 Adhesive test. The adhesive test shall be performed in accordance with ASTM D1583.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or B, or commercial, in unit quantities as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit pack. The unit pack for bags and liners shall be in accordance with Method III of MIL-P-116, in increments of 50 not to exceed 500, in a PPP-B-636, class WR fiberboard box. Bags shall be stacked flat. Bags of extra large size may be folded, provided dummy material is placed at the line of folds to prevent damage. Container closure shall be in accordance with the appendix to PPP-B-636.

5.1.2 Level B.

5.1.2.1 Unit pack. The unit pack for bags and liners shall be as specified in 5.1.1.1 with the following exceptions. The unit pack shall be made in one of the following ways; by placing the bags and liners in a box conforming to PPP-B-566, PPP-B-665, PPP-B-636, Class Domestic, or by wrapping or banding the bags securely.

5.1.3 Commercial. Preservation shall be in accordance with Fed. Std. No. 356.

5.2 Packing. Packing shall be level A or B, or commercial, as specified (see 6.2).

5.2.1 Level A. Unit packs for bags and liners as specified in 5.1.1.1 shall be packed in containers conforming to PPP-B-601, Overseas Type or PPP-B-621, Class 2. Closure shall be in accordance with the appendix to the applicable specification. Whenever practicable, containers shall be of uniform shape and size, of minimum cube and tare and containing identical quantities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. Unit packs for bags and liners as specified in 5.1.2.1 shall be packed in containers conforming to PPP-B-636, Class Domestic or PPP-B-640, Class 1.

5.2.3 Commercial. Packing shall be in accordance with Fed. Std. No. 356.

5.3 Marking.

5.3.1 Level A and B. In addition to any special marking required by the contract (see 6.2), interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129 and as follows:

Specification MIL-B-40028 (including latest revision and amendment)
Size
Month and year of fabrication

5.3.2 Commercial. Marking shall be in accordance with Fed. Std. No. 356.

6. NOTES

6.1 Intended use. The bags covered by this specification are intended primarily for use in packaging of items that are adaptable to protection by volatile corrosion inhibitor treated materials, for indeterminate storage and overseas shipments where high resistance to corrosion of ferrous metal surfaces is required.

6.2 Ordering data. Procurement documents should specify:

a. Title, number and date of this specification.
b. List of drawings, specifications and publications pertinent to the bags and liners showing applicable revision dates.
c. Number of samples required and shipping instructions for first article (see 3.1 and 4.4).
d. Type, class and style of VCI treated liner, if different (see 3.2.2).
e. Bag construction, if different (see 3.3.1).
f. Bag and liner dimensions, if different (see 3.3.3).
g. Inspection lot size, if different (see 4.5.1).
h. Packaging inspection, if different (see 4.5.5).
i. Selection of applicable level of protection (A or B), or commercial packaging (see 5.1, 5.2 and 5.3).
j. Unit pack quantities (see 5.1).
k. Special marking, when required (see 5.3.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.

6.3 Procurement agencies should not purchase more than a one year supply of the bags.

6.4 Bags eighteen months or older should be tested for vapor inhibitors ability in accordance with VIA test of MIL-P-3420, prior to use.

Custodians:
Army - WC
Navy - AS
Air Force - 69

Review activities:
Army - AT, GL
Air Force - 84, 80
Navy - SA

User activities:
Navy - OS, SH
Air Force - 82

Preparing activity:
Army - WC

Project number:
1005-0540

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Rock Island Arsenal
ATTN: SARRI-LE-S
Rock Island, Illinois 61201

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-B-40028B; Bags, Barrier, with Volatile Corrosion Inhibitor Treated Liners

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:02Z
Check the source to verify that this is the current version before use.